How To Write Papers

Dr. Heng Fan, UNT
(some personal experiences)

Speaker: Dr. Yunhe Feng, UNT

July 1, 2026

Partial content/material adapted from Z. Zhou in NJU.

In the Begining

You all are excellent students!



In the Beginning

- A qualified Researcher should be able to independently
  - Come up with ideas
  - Implement ideas
  - Finish paper writing
  - Finish rebuttal
  - Get paper accepted (somewhere good)



What Is The Goal?

- Writing paper is to tell reviewers (and readers):
  - Problem X is important
  - Previous works A, B, and C have been done
  - A, B, and C have their own limitations (e.g., accuracy, efficiency, generality, bias, etc)
  - Your proposed work D (to address the above limitations)
  - Experimental and/or theoretic analysis of your method
  - Comparison with previous works
  - Why your method D is better
  - Strength and weakness of D
  - Future work on D

Paper Structure

- Outline of a paper (conference paper as an example):
  - Title
  - Abstract
  - Introduction
  - Related Work
  - Your Methodology
  - Experiments
  - Conclusion

Paper Structure

- Title
  - Clearly state the key in you paper
  - Don't make it too long (usually less than 15 words)
  - Make it attractive
  - A good title is as important as the abstract and introduction. It is worth time.

Paper Structure

- Title
  - My strategy: attractive words (optional) + key words + purpose words (optional) + task name (note: the order of these words is flexible)
  - Some examples of titles in my papers:
    - Knowing Your Target: Target-Aware Transformer Makes Better Spatio-Temporal Video Grounding (ICLR 2025)
    - LaMOT: Language-Guided Multi-Object Tracking (ICRA 2025)
    - Beyond MOT: Semantic Multi-Object Tracking (ECCV 2024)
    - Tracking Meets LoRA: Faster Training, Larger Model, Stronger Performance (ECCV 2024)
    - Efficient Multimodal Semantic Segmentation via Dual-Prompt Learning (IROS 2024)
    - Context-Guided Spatio-Temporal Video Grounding (CVPR 2024)

Paper Structure

- Abstract
  - Purpose: Summarize your contributions
  - My style (in most cases): writing an abstract by answering questions
    - Q1: What is the problem to solve in this paper?
    - Q2: What is your work? (What will you propose?)
    - Q3: What are the features (keys) of your work?
    - Q4: What are the advantages of your work?
    - Q5: What are the results (may show some numbers)?

Sometimes I may simply start from Q2 (leaving Q1 in the introduction)

Paper Structure

- Abstract
  - Examples:

Transformer has attracted increasing interest in STVG, owing to its end-to-end pipeline and promising result. Existing Transformer-based STVG approaches often leverage a set of object queries, which are initialized simply using zeros and then gradually learn target position information via iterative interactions with multimodal features, for spatial and temporal localization. Despite simplicity, these zero object queries, due to lacking target-specific cues, are hard to learn discriminative target information from interactions with multimodal features in complicated scenarios (e.g., with distractors or occlusion), resulting in degradation. Addressing this, we introduce a novel Target-Aware Transformer for STVG (TA-STVG), which

This part informs the reviewers what problem you want to solve in the paper (i.e., addressing what limitations in current works) – could be long (detailed) or short (abstract)

given
dules,
ASA),
rom a
video utilizing holistic text information, while the latter aims at further exploiting the fine-grained visual attribute information of the object from previous target-aware temporal cues, which is applied for object query initialization. Compared to existing methods leveraging zero-initialized queries, object queries in our TA-STVG, directly generated from a given video-text pair, naturally carry target-specific cues, making them adaptive and better interact with multimodal features for learning more discriminative information to improve STVG. In our experiments on three benchmarks, TA-STVG achieves state-of-the-art performance and significantly outperforms the baseline, validating its efficacy.

Knowing Your Target: Target-Aware Transformer Makes Better Spatio-Temporal Video Grounding, ICLR, 2025.

Paper Structure

- Abstract
  - Examples:

Transformer has attracted increasing interest in STVG, owing to its end-to-end pipeline and promising result. Existing Transformer-based STVG approaches often leverage a set of object

This part tells the reviewers what your method is roughly about (usually 1-2 sentences).

Addressing this, we introduce a novel Target-Aware Transformer for STVG (TA-STVG), which seeks to adaptively generate object queries via exploring target-specific cues from the given video-text pair, for improving STVG. The key lies in two simple yet effective modules, comprising text-guided temporal sampling (TTS) and attribute-aware spatial activation (ASA), working in a cascade. The former focuses on selecting target-relevant temporal cues from a video utilizing holistic text information, while the latter aims at further exploiting the fine-grained visual attribute information of the object from previous target-aware temporal cues, which is applied for object query initialization. Compared to existing methods leveraging zero-initialized queries, object queries in our TA-STVG, directly generated from a given video-text pair, naturally carry target-specific cues, making them adaptive and better interact with multimodal features for learning more discriminative information to improve STVG. In our experiments on three benchmarks, TA-STVG achieves state-of-the-art performance and significantly outperforms the baseline, validating its efficacy.

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This part tells the reviewers what features/keys of your method is (e.g., your proposal modules/strategies/…, usually 2-4 sentences)

seeks to adaptively generate object queries via exploring target-specific cues from the given video-text pair, for improving STVG. The key lies in two simple yet effective modules, comprising text-guided temporal sampling (TTS) and attribute-aware spatial activation (ASA), working in a cascade. The former focuses on selecting target-relevant temporal cues from a video utilizing holistic text information, while the latter aims at further exploiting the fine-grained visual attribute information of the object from previous target-aware temporal cues, which is applied for object query initialization. Compared to existing methods leveraging zero-initialized queries, object queries in our TA-STVG, directly generated from a given video-text pair, naturally carry target-specific cues, making them adaptive and better interact with multimodal features for learning more discriminative information to improve STVG. In our experiments on three benchmarks, TA-STVG achieves state-of-the-art performance and significantly outperforms the baseline, validating its efficacy.

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I usually add this part to emphasize that the importance/advantage of my method – telling reviewers what I can address (for improvements)

grained visual attribute information of the object from previous target-aware temporal cues, which is applied for object query initialization. Compared to existing methods leveraging zero-initialized queries, object queries in our TA-STVG, directly generated from a given video-text pair, naturally carry target-specific cues, making them adaptive and better interact with multimodal features for learning more discriminative information to improve STVG. In our experiments on three benchmarks, TA-STVG achieves state-of-the-art performance and significantly outperforms the baseline, validating its efficacy.

Knowing Your Target: Target-Aware Transformer Makes Better Spatio-Temporal Video Grounding, ICLR, 2025.

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Summarize the experiments (sometimes you can put some numbers here to highlight your method, i.e., setting a new record by achieving 0.76 SUC score, etc)

cues, zero- -text with multimodal features for learning more discriminative information to improve STVG. In our experiments on three benchmarks, TA-STVG achieves state-of-the-art performance and significantly outperforms the baseline, validating its efficacy.

Knowing Your Target: Target-Aware Transformer Makes Better Spatio-Temporal Video Grounding, ICLR, 2025.

Paper Structure

- Abstract
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    - Q5: What are the results (may show some numbers)?

Note 1: My abstract is usually a little long (I prefer detailing the problem to solve). If you want your abstract shorter, you may simply your answers to start from answering Q2.

Note 2: The organization of answers to different questions are different (needing adaption for different papers)

Paper Structure

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Note 3: Abstract is the most important part of your paper. You need to spend a lot of time on this part (iterative revisions/refinements).

I usually complete abstract in the very beginning (some may leave it in the end), because writing abstract is a thinking process of organizing the whole story and helps the writing of subsequent sections.

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - Problem X is important
    - A, B, and C have been done
    - A, B, and C are limited in accuracy/efficiency/…
    - The inspiration (optional) and your proposed D
    - Features/keys and advantages of D
    - Results
    - Summary of contributions

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - Problem X is important
      - ❖ Introduce what X is; what X is used for; why do we study X
      - ❖ one paragraph

Spatio-temporal video grounding (or ***STVG***) aims at *spatially* and *temporally* localizing the target of interest from an untrimmed video given its textual description (Zhang et al., 2020b). As a multimodal task, it requires to understand the spatio-temporal content from videos and accurately connect it to the corresponding textual expression. Owing to its importance in multimodal video understanding and key applications such as content-based video retrieval, robotics, etc, STVG has attracted increasing interest with many models proposed (Zhang et al., 2020a;b; Tang et al., 2021; Su et al., 2021).

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - A, B, and C have been done
    - A, B, and C are limited in accuracy/efficiency/…
      - Introduce what other works are about; what are their limitations and why they are limited; what are the consequences due to limitations
      - There might be a lot of prior works. In introduction, you only need to list the very relevant ones
      - When discussion the limitations, the limitations must be related to what you will address
      - One paragraph (a little longer maybe – L+)

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - A, B, and C have been done
    - A, B, and C are limited in accuracy/efficiency/…

Recently, inspired by the compact end-to-end pipeline of Transformer-based detection (Carion et al., 2020), researchers have introduced the Transformer (Vaswani et al., 2017) for STVG given that they both are localization task, and achieved previously unattainable result (Yang et al., 2022a; Jin et al., 2022; Lin et al., 2023b; Gu et al., 2024). Similar to Transformer-based detection (Carion et al., 2020), existing Transformer-based STVG methods often adopt a set of spatial and temporal object queries, and then leverage them to learn target position information by iterative interactions with multimodal features (generated by the encoder) from video and text for spatio-temporal target localization. In these approaches (Yang et al., 2022a; Jin et al., 2022; Lin et al., 2023b; Gu et al., 2024), the object queries are usually simply initialized using *zeros* (see Fig. 1 (a)). Despite promising results, such zero-initialized object queries, due to the lack of effective target-specific semantic cues, are difficult to learn discriminative target position information from the interactions with multimodal features in complicated scenarios, *e.g.*, with distractors or occlusion, leading to performance degradation.

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
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information from the interactions with mu
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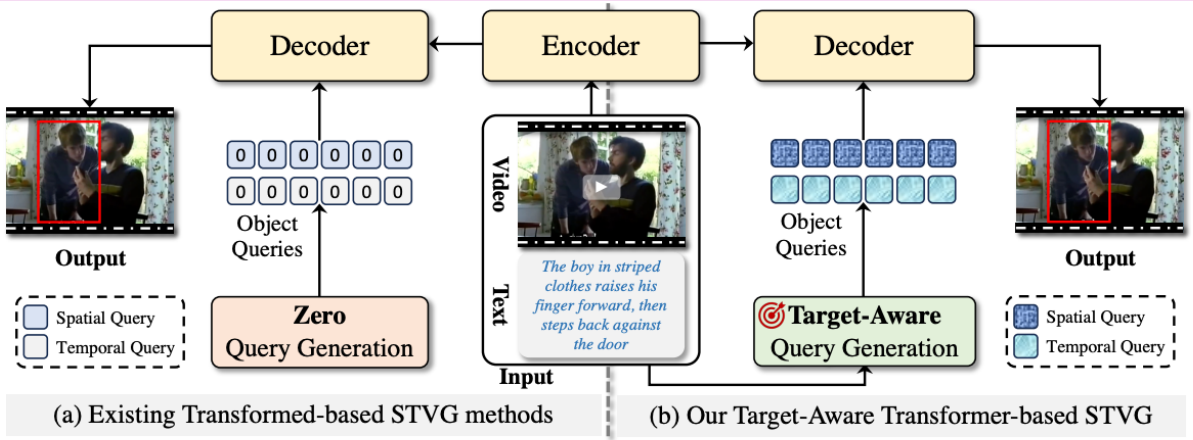


Figure 1: Comparison between existing Transformer-based STVG methods applying zero-initialized queries for STVG in (a) and our proposed Target-Aware Transformer-based STVG generating queries with target-aware cues from video and text for STVG in (b). *Best viewed in color for all figures.*

I like adding a figure to highlight the difference between prior works and mine

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - The inspiration (optional) – I prefer having it in my paper
      - Let reviewers know how you get or why you propose the idea D.
      - Note, this part is not about the details of D, b, but a transition stage to your proposed work D.
      - One paragraph
    - Your proposed D
      - Detail the features, keys and advantages of your proposed D
      - One paragraph (a little bit longer)

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - The inspiration (optional) – I prefer having it in my paper

In the decoder procedure, object queries expect to learn target position information from multimodal features. If *object queries know the target from the very beginning*, or in other words, *they know what to learn*, they can employ target-specific cues as a prior to guide themselves for better interaction with the multimodal features, which benefits learning more discriminative features for better localization. In order to validate this, we conduct the *oracle* experiment by generating spatial and temporal object queries from groundtruth, and compare the performance using such groundtruth-generated queries and zero-initialized queries. The architecture in this oracle experiment simply consists of feature extraction, an encoder and a decoder, and please refer to the supplementary material for details due to limited space. Fig. 2 shows the comparison results on HCSTVG-v1 (Tang et al., 2021), and more comparison on other benchmarks can be found in Sec. A in the supplementary material. From Fig. 2, we can clearly observe that, when introducing the target-specific information from groundtruth to initialize object queries, the STVG performance can be significantly improved, *e.g.*, from 49.9% to 68.9% in m_IoU (18.0% absolute gains) and 36.4% to 49.9% in m_vIoU (13.5% absolute gains), fully verifying effectiveness of target-specific information in initializing queries for accurate STVG.

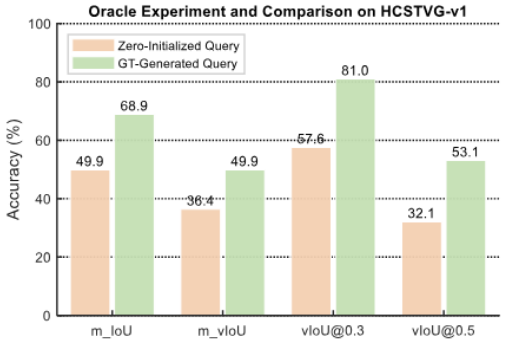


Figure 2: Comparison of the zero-initialized queries and groundtruth-generated queries for STVG. We see the target-specific information in groundtruth largely enhances results.

In this inspiration paragraph, I use experiment to tell reviewers a finding, which introduces the following proposed method

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - The inspiration (optional) – I prefer having it in my paper
    - Your proposed D

Thus motivated by the above, we introduce a novel ***Target-Aware Transformer for STVG*** (TA-STVG), which seeks to adaptively initialize object queries by exploring target-specific cues from videos for spatio-temporal target localization (see Fig. 1 (b)). The crux of our TA-STVG lies in two simple yet effective modules, including *text-guided temporal sampling* (TTS) and *attribute-aware spatial activation* (ASA). These two modules work in a cascade in the TA-STVG, from coarse to fine, to generate the target-aware queries for subsequent localization. Specifically, TTS, from the temporal perspective, firstly aims to select target-relevant frames from the video using holistic information from the textual expression. To enhance the temporal selection, both appearance and motion cues are jointly considered in TTS. Then, from a spatial perspective, ASA focuses on exploring fine-grained visual semantic information of the object from the previous target-aware temporal cues learned by TTS. Particularly, attributes such as color and action will be exploited in the ASA to activate spatial features most relevant to the target object. Considering the specificity of object queries for spatial and temporal decoders, different activation features based on the specific attributes will be generated. Finally, features from ASA are used to initialize queries, which are fed to decoders, interacting with multimodal features from the encoder, to learn target position information for localization.

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - Results
      - ❖ Your proposed D is superior to other approaches
      - ❖ One paragraph

In this work, we employ the DETR-similar architecture for TA-STVG, similar to existing Transformer-based STVG methods (Yang et al., 2022a; Jin et al., 2022; Lin et al., 2023b; Gu et al., 2024), but ***difference*** is that, our object queries, for both spatial and temporal localization, are directly generated from the given video-text pair, and thus naturally carry the target-specific cues, making them adaptive and better interact with multimodal features for learning more discriminative information to improve STVG. To our best knowledge, TA-STVG is the first to incorporate target-specific cues for object query generation to improve STVG. To validate its effectiveness, we conduct experiments on three challenging benchmarks, including HCSTVG-v1/-v2 (Tang et al., 2021) and VidSTG (Zhang et al., 2020b), and the results show that TA-STVG outperforms previous Transformer-based STVG methods by achieving new state-of-the-arts, evidencing efficacy of our proposed method for STVG. Moreover, our proposed TTS and ASA modules are designed for general purpose and thus applicable to other architectures. We apply TTS and ASA on top of two approaches, including the seminal Transformer-based TubeDETR (Yang et al., 2022a) and STCAT (Jin et al., 2022), significantly improving their performance, as demonstrated in our experiments, further highlighting the generality of our approach.

Paper Structure

- Introduction
  - Purpose: Background and organization of your story
  - My style: introduction as an expansion of abstract with additional details and context
    - Summary of contributions
      - ❖ Highlight your major contributions
      - ❖ One paragraph

In summary, the major contributions of this work are as follows: ♠ We present the TA-STVG, a novel Target-Aware Transformer for improving STVG by exploring target-specific cues for object queries; ♥ We propose text-guided temporal sampling (TTS) for selecting target-relevant temporal cues from the videos; ♣ We present attribute-aware spatial activation (ASA) to exploit fine-grained visual semantic attribute information for object query generation; and ♦ TA-STVG in extensive experiments achieves new state-of-the-art performance and shows good generality, demonstrating its efficacy.

Note: Introduction is as important as abstract. You need to spend a lot of time on this part.

Paper Structure

- Related Work
  - Purpose: Introduce relevant works and illustrate your uniqueness and difference
  - My style:
    - Categorization of previous works
    - Discuss the strength and weakness of each type of methods
    - Highlight your different in the end
    - 3-4 paragraphs (each paragraph discuss one line of works)

Spatio-Temporal Video Grounding. Spatio-temporal video grounding (STVG) (Zhang et al., 2020b) aims to spatially and temporally localize the target of interest, given its free-form of textual description, from an untrimmed sequence. Early STVG approaches (Zhang et al., 2020b;a; Tang et al., 2021) mainly consists of two stages, *first* generating candidate proposals from the video with a pre-trained detector and *then* selecting correct proposals based on the textual expression. To eliminate the heavy dependency on the pre-trained detection model, recent methods (Su et al., 2021; Yang et al., 2022a; Jin et al., 2022; Lin et al., 2023b; Talal Wasim et al., 2024; Gu et al., 2024), inspired by Transformer, switch to the one-stage pipeline that directly generates a spatio-temporal tube for target localization, without relying on any external detectors. Owing to the compact end-to-end training pipeline, such a one-stage framework demonstrates superior performance compared to previous two-stage algorithms. Our TA-STVG also belongs to the one-stage Transformer-based type. However, ***different from*** the aforementioned Transformer-based approaches that simply follows (Carion et al., 2020) to leverage zero-initialized object queries for target localization, TA-STVG innovatively exploits target-specific cues from the video-text pair for object query generation, making it adaptive to various scenarios and better interact with multimodal features in the decoder for more accurate localization.

Example of one of paragraphs

I like adding a part to emphasize the difference between other works and mine. This helps reinforce the advantage and uniqueness of your method.

Paper Structure

- Your Methodology
  - Purpose: Introduce your work
  - There is no single, set style. My style is usually as follows:
    - An overview paragraph introducing the pipeline (corresponding to an overview figure) – one paragraph
    - Main part – strategy 1
      - First introducing the overall framework (could be one subsection, or multiple subsections – depending on your specific method), where your proposed modules (give them cool names) are leveraged
      - Then detail the proposed modules separately after the framework
      - Sometimes you may want to add a training objective (loss function) of your framework (this part can be put in supplementary if space is limited)

Paper Structure

- Your Methodology
  - Purpose: Introduce your work
  - There is no single, set style. My style is usually as follows:
    - An overview paragraph introducing the pipeline (corresponding to an overview figure) – one paragraph
    - Main part – strategy 2
      - Based on the logic of your proposed method, sequentially introducing each of the component
      - Sometimes you may want to add a training objective (loss function) of your framework (this part can be put in supplementary if space is limited)

The adoption of strategy 1 or 2 depends the specific methodology. The important thing is to **make your method clear to reviewers and readers**.

Paper Structure

- Your Methodology
  - Purpose: Introduce your work
  - There is no single, set style. My style is usually as follows:

3 Our Approach

Overview. In this work, we propose TA-STVG, which aim
improving STVG. Motivated by DETR (Carion et al., 20
framework which consists of a multimodal encoder and
shown in Fig. 3, the multimodal encoder (§3.1) extracts ar
the decoder (§3.4) learns target position information using
TTS (§3.2) and ASA (§3.3) and their interaction with the

3.1 Multimodal Encoder

3.2 Text-Guided Temporal Sampling (TTS)

3.3 Attribute-aware Spatial Activation (ASA)

3.4 Decoder for Spatio-Temporal Localization

3.5 Optimization

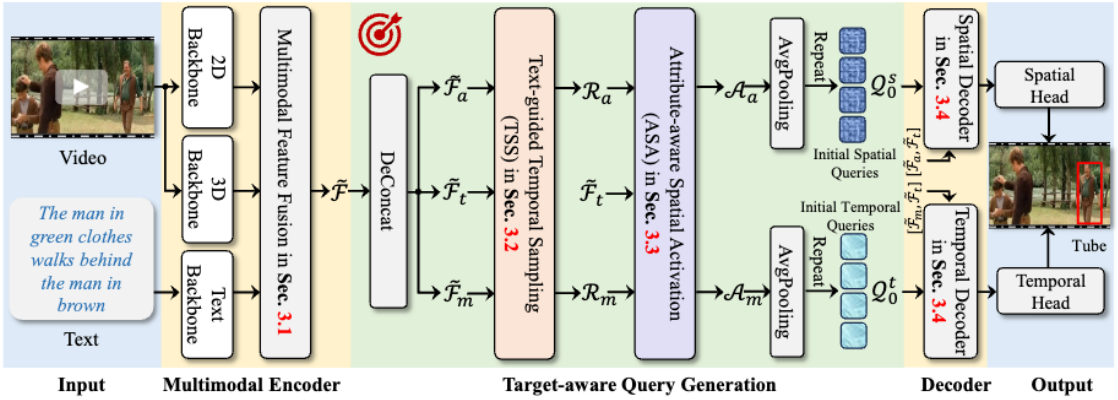


Figure 3: Overview of TA-STVG, which exploits target-specific information from the video and text (*i.e.*, features from multimodal encoder) for generating spatial and temporal object queries for STVG.

Strategy 2: sequentially introducing each component

Paper Structure

- Your Methodology
  - Purpose: Introduce your work
  - There is no single, set style. My style is usually as follows:

3. The Proposed Method

Overview. In this work, we introd
the discriminative visual context
guidance to improve the target loc
DETR [5] for detection, CG-ST
decoder architecture, which comp
coder (Sec. 3.1) and the context-g
As illustrated in Fig. 2, the encode
timodal visual-linguistic feature t
mation from text query, which is
decoder for target localization gu
learned with ICG (Sec. 3.3) and IC

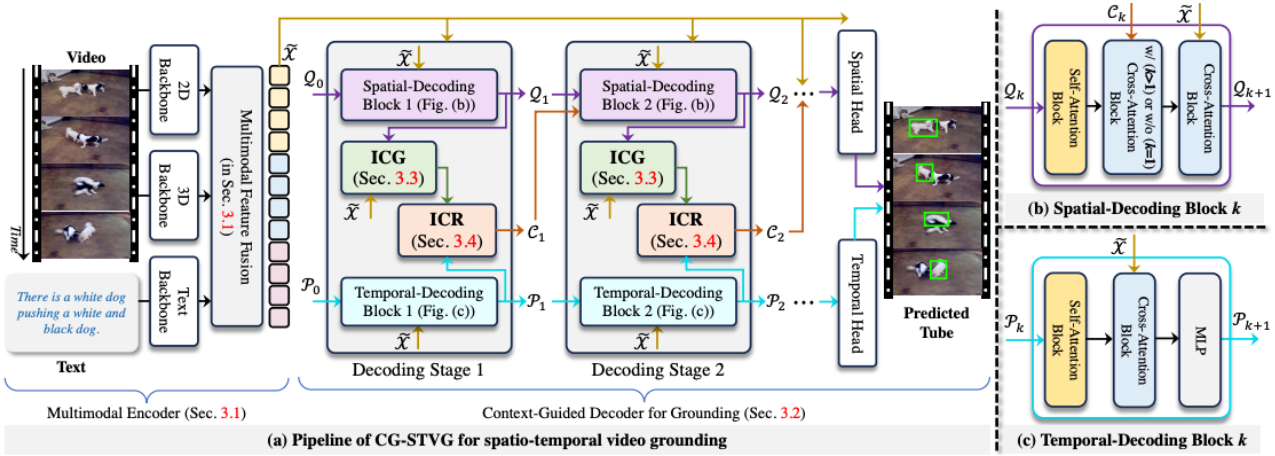


Figure 2. Overview of our method, which consists of a multimodal encoder for feature extraction and a context-guided decoder by cascading a set of decoding stages for grounding. In each decoding stage, instance context is mined to guide query learning for better localization.

3.1. Multimodal Encoder
3.2. Context-Guided Decoder for Grounding

framework

3.3. Instance Context Generation (ICG)
3.4. Instance Context Refinement (ICR)

proposed modules

3.5. Optimization

Strategy 1: introducing overall framework, then detail proposed modules later

Paper Structure

- Experiments
  - Purpose: Experimental support to your method
  - My style is usually as follows:
    - A paragraph for implementation details.
    - Subsection 1: One or two paragraphs to describe used datasets and evaluation metrics. (this may not necessarily be a subsection; it's up to you)
    - Subsection 2: state-of-the-art comparison to other methods.
    - Subsection 3: Ablation studies of different designs.
    - Subsection 4: Discussion/analysis (e.g., efficiency, qualitative results, generalization of your method, etc). The discussion of limitation (if there is) is put in the appendix.

Paper Structure

- Conclusions
  - Purpose: Summary and future work
  - My style is usually as follows:
    - Summary.
    - Future work. (sometimes the future work is omitted; but you are suggested to add this for completeness)

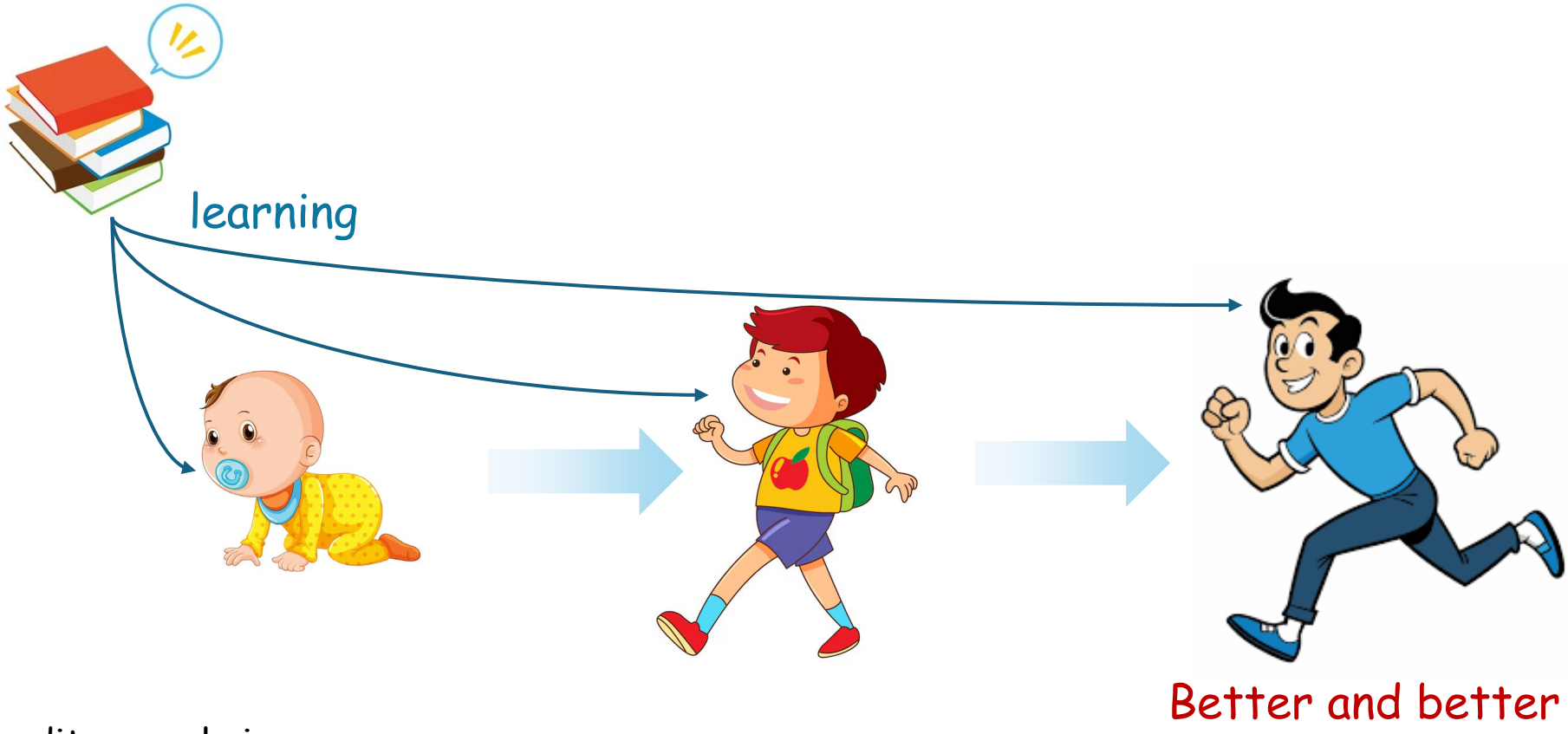
Paper Structure

- Conclusions
  - Purpose: Summary and future work
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    - Future work. (sometimes the future work is omitted; but you are suggested to add this for completeness)

- Acknowledgement
- Reference
- Appendix (or supplementary material): put tedious details and additional results/analysis in Appendix

Some Suggestions/Tricks

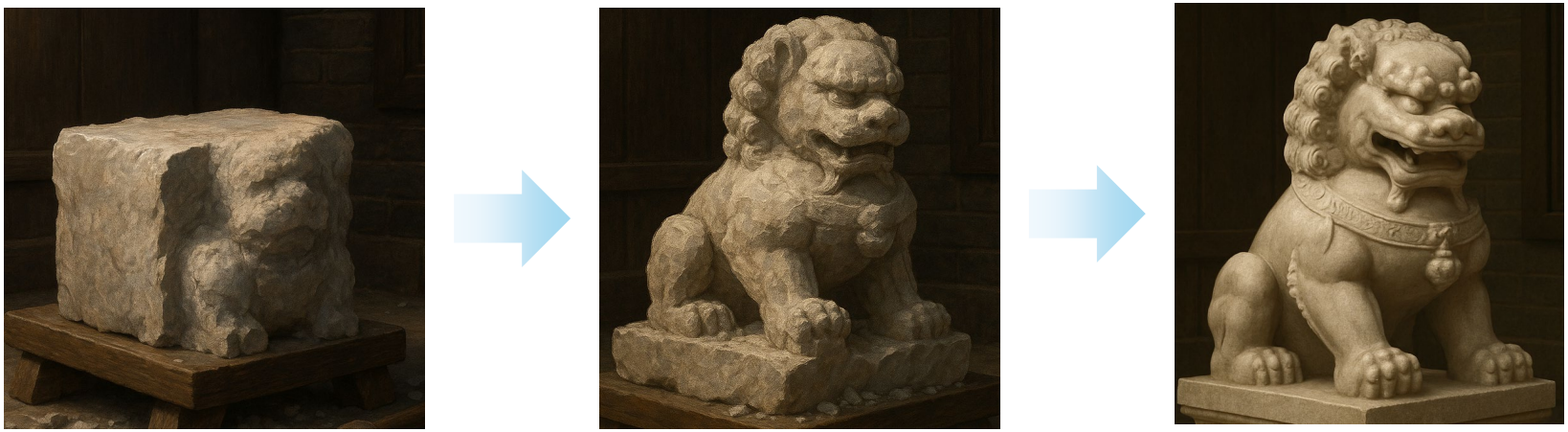
- Learning from other excellent papers (published in top-tier conferences/journals)
  - Learning not just idea, but also writing



Credits: google images

Some Suggestions/Tricks

- Revise, revise, and revise
  - Writing is not a one-time thing; you need to iteratively revise back and forth;
  - Discuss with your advisor/collaborators, and improve from comments
  - Start writing as soon as possible; give some time to your advisor and collaborators



Credits: Chatgpt-generated images

Some Suggestions/Tricks

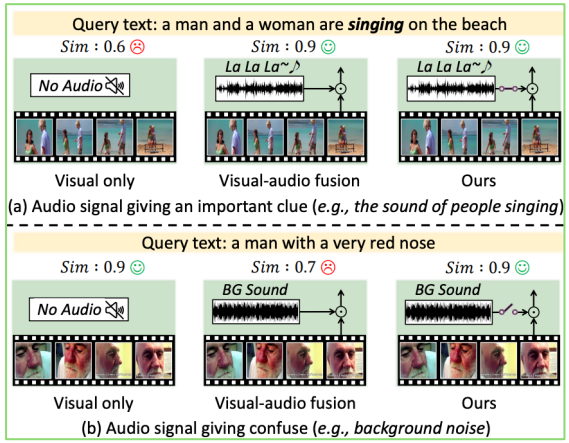
- Try to use simple sentences
  - Scientific writing is not like novel or poem writing; The most important thing is to make every point clear so reviewers and readers can understand it easily.
- Write every word on your own
  - Do NOT copy from other papers (even your own papers). You can rephrase similar statements.
  - My opinion: Use tools like ChatGPT only when you know how to write papers; don't use it when you start the learning of writing

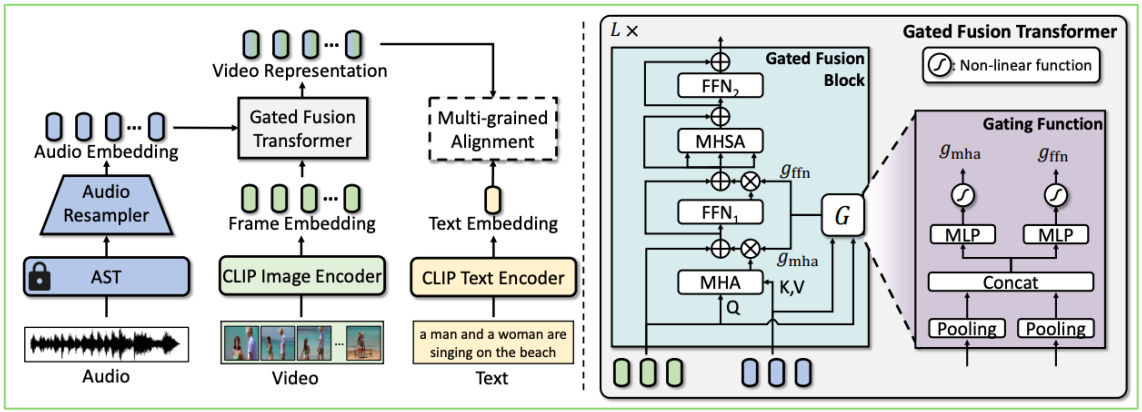
Some Suggestions/Tricks

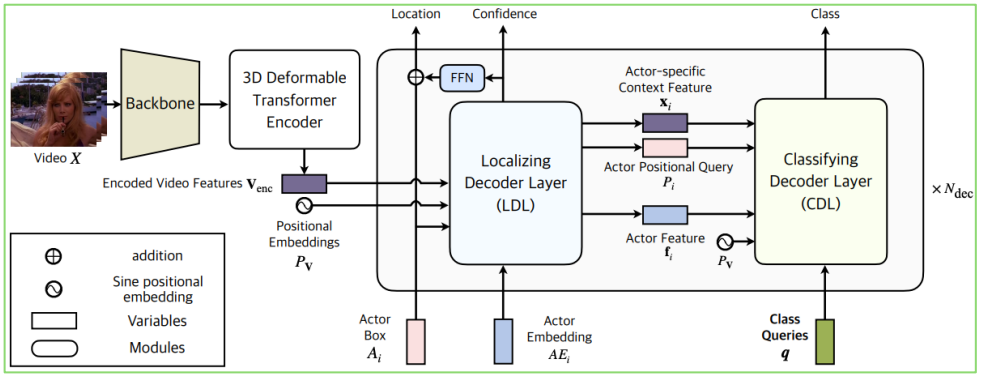
- Try to make good-looking figures
  - These are what you can control in your paper, so try to make them good and give reviewers a good impression
  - Learn from other papers
  - Tool: Power Point is good enough to me

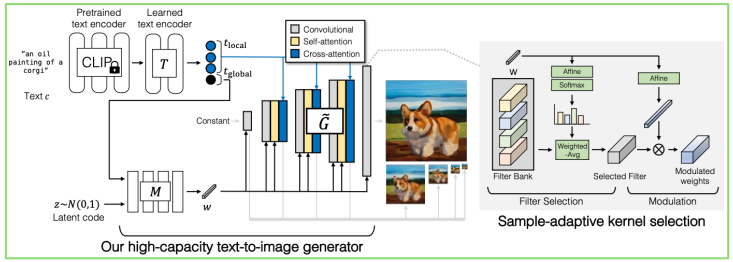
Some Suggestions/Tricks

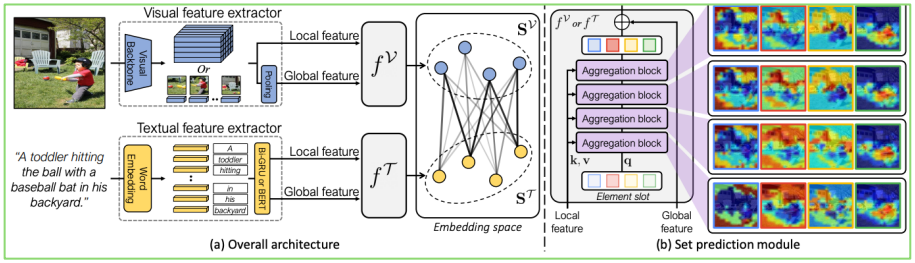
- Try to make good-looking figures
  - Some examples (not from my paper)











Some Suggestions/Tricks

- Try to conduct extensive experiments/analysis
  - Extensive and solid experiments are crucial (in reviewers' eye) to justify the effectiveness of your work
  - Rich ablations are required for analysis
  - Tables in experiments should be compact and elegant
  - Highlight your results in tables
  - Improve readability of your tables in ablations (adding orders)

Some Suggestions/Tricks

- Try to conduct extensive experiments/analysis

Table 1: Comparison on HCSTVG-v1 (%).

Methods	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
STVGBert (Su et al., 2021)	-	20.4	29.4	11.3
TubeDETR (Yang et al., 2022a)	43.7	32.4	49.8	23.5
STCAT (Jin et al., 2022)	49.4	35.1	57.7	30.1
SGFDN (Wang et al., 2023c)	46.9	35.8	56.3	37.1
STVGFormer (Lin et al., 2023b)	-	36.9	62.2	34.8
CG-STVG (Gu et al., 2024a)	52.8	38.4	61.5	36.3
Baseline (ours)	49.9	36.4	57.6	32.1
TA-STVG (ours)	**53.0** (+3.1)	**39.1** (+2.7)	**63.1** (+5.5)	**36.8** (+4.7)

Table 2: Comparison on HCSTVG-v2 (%).

Methods	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
PCC (Yu et al., 2021)	-	30.0	-	-
2D-Tan (Tan et al., 2021)	-	30.4	50.4	18.8
MMN (Wang et al., 2022)	-	30.3	49.0	25.6
TubeDETR (Yang et al., 2022a)	53.9	36.4	58.8	30.6
STVGFormer (Lin et al., 2023b)	58.1	38.7	65.5	33.8
CG-STVG (Gu et al., 2024a)	60.0	39.5	64.5	36.3
Baseline (ours)	58.3	37.9	62.6	33.0
TA-STVG (ours)	**60.4** (+2.1)	**40.2** (+2.3)	**65.8** (+3.2)	**36.7** (+3.7)

Table 3: Comparison with existing state-of-the-art methods on VidSTG (%).

Methods	Declarative Sentences				Interrogative Sentences			
	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
STGRN (Zhang et al., 2020b)	48.5	19.8	25.8	14.6	47.0	18.3	21.1	12.8
OMRN (Zhang et al., 2020a)	50.7	23.1	32.6	16.4	49.2	20.6	28.4	14.1
STGVT (Tang et al., 2021)	-	21.6	29.8	18.9	-	-	-	-
STVGBert (Su et al., 2021)	-	24.0	30.9	18.4	-	22.5	26.0	16.0
TubeDETR (Yang et al., 2022a)	48.1	30.4	42.5	28.2	46.9	25.7	35.7	23.2
STCAT (Jin et al., 2022)	50.8	33.1	46.2	32.6	49.7	28.2	39.2	26.6
SGFDN (Wang et al., 2023c)	45.1	28.3	41.7	29.1	44.8	25.8	36.9	23.9
STVGFormer (Lin et al., 2023b)	-	33.7	47.2	32.8	-	28.5	39.9	26.2
CG-STVG (Gu et al., 2024a)	51.4	34.0	47.7	33.1	49.9	29.0	40.5	27.5
Baseline (ours)	49.5	32.3	44.9	31.6	48.6	27.5	38.5	25.6
TA-STVG (ours)	**51.7** (+2.2)	**34.4** (+2.1)	**48.2** (+3.3)	**33.5** (+1.9)	**50.2** (+1.6)	**29.5** (+2.0)	**41.5** (+3.0)	**28.0** (+2.4)

Some Suggestions/Tricks

- Try to conduct extensive experiments/analysis

Impact of TTS and ASA. The TTS and ASA are key components of TA-STVG. To validate their effectiveness, we conducted ablation experiments on HCSTVG-v1. As shown in Tab. 4, without TTS and ASA (❶), our baseline achieves a m_tIoU score of 49.9%. When TTS is applied alone, the m_tIoU score is improved 52.2% with 2.3% gains (❶ *v.s.* ❷). When ASA is used alone, m_tIoU is improved to 51.4% with 1.5% gains (❶ *v.s.* ❸). Working together, the best result of 53.0% is achieved with 3.1% absolute gains (❶ *v.s.* ❸). All these results clearly validate the efficacy of TTS and ASA, working either alone or together, for improving STVG.

Table 4: Ablations of TTS and ASA.

	TTS	ASA	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
❶	-	-	49.9	36.4	57.6	32.1
❷	✓	-	52.2	38.4	61.7	36.2
❸	-	✓	51.4	38.0	60.4	34.1
❹	✓	✓	**53.0**	**39.1**	**63.1**	**36.8**

Table 5: Ablations of branches in TTS. "TG", "AB", and "MB" are the text-guided, appearance and motion branches, respectively.

	TG	AB	MB	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
❶	-	-	-	51.4	38.0	60.4	34.1
❷	✓	✓	-	51.8	38.4	61.1	36.3
❸	✓	-	✓	52.3	38.3	62.0	36.5
❹	-	✓	✓	51.8	38.5	62.0	36.9
❺	✓	✓	✓	**53.0**	**39.1**	**63.1**	**36.8**

Table 6: Ablations of attributes in ASA. "SG", "AA", and "MA" are the subject-guided, appearance and motion attributes, respectively.

	SG	AA	MA	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
❶	-	-	-	52.2	38.4	61.7	36.2
❷	✓	✓	-	52.3	38.6	62.4	36.2
❸	✓	-	✓	52.7	38.6	61.3	36.6
❹	-	✓	✓	52.6	38.8	61.9	**36.8**
❺	✓	✓	✓	**53.0**	**39.1**	**63.1**	**36.8**

Table 7: Ablations of activation learning in ASA.

	Activation	m_tIoU	m_vIoU	vIoU@0.3	vIoU@0.5
❶	None	52.2	38.4	61.7	36.2
❷	Instance-act.	52.8	38.8	62.0	34.7
❸	Attribute-act.	**53.0**	**39.1**	**63.1**	**36.8**

Table 8: Ablation of δ.

	δ	m_tIoU	m_vIoU
❶	0.4	52.7	38.8
❷	0.5	**53.0**	**39.1**
❸	0.6	52.8	39.0

Table 9: Ablation of θ.

	θ	m_tIoU	m_vIoU
❶	0.6	52.3	38.5
❷	0.7	**53.0**	**39.1**
❸	0.8	52.6	38.6